# OMNIBRIDGE POS™

## Universal Rust Gateway for POS Hardware, Weighing Scales & Printers

■ NATIVE RUST AGENT ■ UNIFIED PORT 8989 ■■ ZERO BROWSER DIALOGS ■ ED25519 OFFLINE LICENSING

**Connect web and cloud POS applications to physical checkout hardware via local HTTP REST endpoints.** OmniBridge POS bypasses browser sandbox limitations, empowering React, Angular, Vue, PHP, Odoo, or WooCommerce systems to operate thermal receipt printers, electronic scales, and cash drawers natively, reliably, and unattended.

**THE HARDWARE CHALLENGE IN WEB & CLOUD APPLICATIONS:**
Modern browsers block direct access to COM serial ports and force intrusive print dialogs (`Ctrl+P`), slowing down checkout speed and requiring cashier interaction for every printed receipt.
**THE OMNIBRIDGE POS SOLUTION:** An ultra-lightweight native Rust daemon running in the background of each POS terminal. It exposes a unified REST API on port 8989 to dispatch thermal receipts in milliseconds, read serial scales in real time, and kick cash drawers without popups or internet dependencies.

| Your Web POS / Cloud ERP | ■■■■■■( Local REST HTTP :8989 )■■■■■■■ | OmniBridge POS (Rust) | ■■ ■ | Checkout Peripherals |
|---|---|---|---|---|
| React • Vue • Odoo • PHP • Woo | PNA / Strict CORS • JSON & PDF | Direct Spooler • Serial COM • RJ11 | | Receipts • Scales • Drawers • Labels |

| < 15 ms | < 15 MB | Auto-COM | Offline Ready |
|---|---|---|---|
| Local response latency | Ultra-low RAM usage | Serial scale auto-scan | Ed25519 crypto licensing |

### Supported Hardware Capabilities & Implemented Flows

| | |
|---|---|
| **Weighing Scales**<br>**RS232 / USB Serial** | Automatic candidate port discovery, buffer clearance, and real-time decimal weight extraction (kg/lb). Includes offline simulation mode for testing. |
| **Receipt Printing**<br>**Thermal ESC/POS** | Silent unattended PDF and RAW ESC/POS job dispatch directly to the OS print spooler. Automatic Windows printer-copy name normalization. |
| **Barcode Labels**<br>**TSPL / ZPL / RAW** | Direct RAW path for shelf labeling and barcode generation. Accepts standard thermal printer commands. |
| **Continuous Form**<br>**Dot Matrix ESC/P** | Plain-text and ESC/P escape sequences with automatic page eject (Form Feed) for industrial dot-matrix billing printers. |
| **Cash Drawer & Docs**<br>**RJ11 / Office PDF** | Instant drawer kick-out pulses via RJ11/RJ12 printer ports (Pin 2 and Pin 5). Full-page A4/Letter PDF printing with proportional fitting. |

### Quick 3-Line Integration (Web POS / Frontend)

**LOCAL INTEGRATION ARCHITECTURE (Port 8989):**
• **1-Click Setup:** Runs as a Windows Service, Linux Systemd daemon, or macOS Launchd agent.
• **Web Compliant:** Fully supports Chrome Private Network Access (PNA) and strict CORS origin allowlisting.
• **Built-in Dashboard:** Interactive diagnostic web panel running at http://localhost:8989/.
• **Zero Cloud Dependency:** Checkout operations execute entirely offline.

**FRONTEND JAVASCRIPT CONSUMPTION:**

```
// 1. Read real-time scale weight:
const { peso } = await (await fetch('http://localhost:8989/peso')).json();
// 2. Trigger cash drawer kick pulse:
await fetch('http://localhost:8989/drawer', { method: 'POST' });
// 3. Print silent thermal receipt in milliseconds:
await fetch('http://localhost:8989/ticket', {
  method: 'POST', headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify({ content: "RECEIPT #101\nTotal: $25.00\n" })
});
```

## Core API Endpoints & Real-Time Diagnostics

| Method | REST Route | Purpose & Action | Payload / Response |
|---|---|---|---|
| **GET** | / | Interactive visual web dashboard with 1-click test suite | HTML Dashboard |
| **GET** | /peso, /weight | Real-time physical weight read from serial scale | JSON {peso: 1.25} |
| **POST** | /ticket | Silent printing to configured thermal receipt printer | JSON / PDF / RAW |
| **POST** | /drawer | Trigger electric cash drawer kick pulse (RJ11) | Pulse RJ11 Pin 2/5 |
| **POST** | /print-label | Direct RAW barcode label dispatch (TSPL/ZPL) | JSON {commands} |
| **POST** | /printers/cleanup | Auto-purge and spooler recovery for jammed queues | Spooler Recovery |
| **GET** | /health, /status | Complete health check across ports, spooler & licenses | JSON Health Status |

## Technical Specifications & Hardware Validation Matrix

| Component | Rust Implementation | Performance / Protocol | Deployment & Validation |
|---|---|---|---|
| **Core Engine** | Native Rust 2021 | < 15 MB RAM, sub-second | Standalone single binary |
| **Supported OS** | Cross-platform | Windows 10/11, Linux, macOS | Windows / Systemd / Launchd |
| **Security** | Offline Ed25519 | Asymmetric crypto signatures | Chrome PNA & Strict CORS |
| **Hardware** | Serial & Spooler | COM auto-scan & spool heal | Pilot test recommended |

## Commercial Licensing & Integration Options

**Per-Station / POS License**
**Retail Stores & Supermarkets**
**$35 USD / year** (or $90 lifetime)

- 1-Click installer running as an automatic Windows Service.
- Per-station license (Annual or Lifetime Perpetual).
- **Local diagnostic dashboard** (:8989) with auto-healing spooler.
- Unlimited scale reading and receipt printing with zero watermarks.
- Dedicated onboarding and setup support.

**OEM / White-Label Partner**
**Software Houses & ERP SaaS**
**$1,490 USD** (One-time fee)

- Unlimited distribution under your own brand name and logo.
- Complete **Postman** collection and **OpenAPI 3.0** spec.
- Zero per-seat recurring royalties.
- Dedicated license generator tool to issue keys to your own clients.
- Ready-to-paste integration snippets for your web ERP.

**Custom Engineering & Drivers**
**Bespoke Hardware Integrations**
**Bespoke Project Quote**

- Custom industrial scale protocols and serial driver tuning.
- Assisted integration with Odoo, WooCommerce, PHP, React, or Python.
- Dedicated guidance during pilot deployments and field tests.
- Custom store printing workflow adaptations.
- Priority direct engineering support.

### READY TO INTEGRATE HARDWARE INTO YOUR WEB POS?

Download the functional DEMO installer or request a bespoke commercial quote:

**WhatsApp / Phone:** +58 412-6165227 | **Email:** drinowar@gmail.com
**Lead Developer:** Darwin Padrino | **Official DEMO Releases:** github.com/dpadrino/omnibridge-pos-demo/releases